CMSC250 Quiz 1

Discrete Mathematics | Instructor Saurav Patel

UID (9 digits): 1 2 3 4 5 6 7 8 9

Student Name: Will Dunbar

Directory ID: wdunbar07

Instructions: Answer all questions. Show your reasoning where requested. For multiple-choice questions, circle one answer.

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1. Truth Table (6 points). Complete the truth table for the proposition $(p \wedge \neg q) \vee q$. Use T for true and F for false.

p	q	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \vee q$
T	T	F	F	T
T	F	T	T	T
F	T	F	F	T
F	F	T	F	F

2. Multiple Choice (2 points). Which expression is logically equivalent to $\neg(p \vee q)$?

A. $\neg p \vee \neg q$

(B.) $\neg p \wedge \neg q$

C. $p \wedge q$

D. $p \vee \neg q$

3. Multiple Choice (2 points). Which statement is the contrapositive of "If n is divisible by 4, then n is even"?

A. If n is even, then n is divisible by 4.

B. If n is not divisible by 4, then n is not even.

(C.) If n is not even, then n is not divisible by 4.

D. If n is divisible by 4, then n is not even.

4. Short Answer (4 points). Define what it means for two propositions to be *logically equivalent*. Give one example of two logically equivalent propositions.

The statement is true if they have the same truth under every possible assignment of their variables

5. Integer Proof (6 points). Prove that for every integer n, the integer $n^2 + n$ is even. Clearly state the definition of an even integer and justify your steps.

An even integer can be written as $2k \ \forall \ k \in \mathbb{Z}$.

$n^2 + n \equiv n(n+1)$ and $n, n+1$ are consecutive so one of them is even

Since either n or $(n+1)$ is even, it is a multiple of 2, so $n(n+1) \approx 2(j), \ j \in \mathbb{Z}$ ∎

CMSC250 Quiz 1

Discrete Mathematics | Instructor Saurav Patel

UID (9 digits): | | | | | | | | | |

Student Name: Rachel Washington

Directory ID: rswash04

Instructions: Answer all questions. Show your reasoning where requested. For multiple-choice questions, circle one answer.

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D. If n is divisible by 4, then n is not even.

4. Short Answer (4 points). Define what it means for two propositions to be *logically equivalent.* Give one example of two logically equivalent propositions.

For all combinations of statements, we should assign all possible combination values. When we recieve the same outcome for each expression, they are logically equivalent statements.

5. Integer Proof (6 points). Prove that for every integer n, the integer $n^2 + n$ is even. Clearly state the definition of an even integer and justify your steps.

~~I wrote some stuff~~

~~Think I got some where ∀ → a person~~

~~nevermind need more space must scratch off~~

SEE scratch page!

⟹

Rachel Washington

⑤ Lets say n is odd. So $n = 2k+1$, $k \in \mathbb{Z}$

$$\Rightarrow n^2+n = (2k+1)^2 + 2k+1 \quad \leftarrow \text{substitution}$$

$$= 4k^2+4k+1+2k+1 \quad \leftarrow \text{expansion}$$

$$= 2(2k^2+3k+1) \quad \leftarrow \text{Factor out } 2$$

since $2k^2+3k+1 \in \mathbb{Z}$, $2(2k^2+3k+1)$ is even by definition

$\therefore n^2+n$ is even ▨

CMSC250 Quiz 1

Discrete Mathematics | Instructor Saurav Patel

UID (9 digits): 120107413

Student Name: Ani Sura

Directory ID: Asur

Instructions: Answer all questions. Show your reasoning where requested. For multiple-choice questions, circle one answer.

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D. If n is divisible by 4, then n is not even.

4. Short Answer (4 points). Define what it means for two propositions to be *logically equivalent.* Give one example of two logically equivalent propositions.

They have the same meaning and can be derived from each other in a proof.

5. Integer Proof (6 points). Prove that for every integer n, the integer $n^2 + n$ is even. Clearly state the definition of an even integer and justify your steps.

Lets say n is odd. Then $n = 2k+1$, $k \in \mathbb{Z}$. $(2k+1)^2 + (2k+1)$

$= 4k^2 + 4k + 1 + 2k + 1 = 4k^2 + 6k + 2 = 2(2k^2 + 3k + 1) \rightarrow$ even ✓

Say n is even so $n = 2j$. $\Rightarrow (2j)^2 + 2j = 4j^2 + 2j = 2(2j^2 + j) \rightarrow$ even ✓

$\therefore n^2 + n$ is even ∎

CMSC250 Quiz 1

Discrete Mathematics | Instructor Saurav Patel

UID (9 digits): 1 2 0 4 2 7 2 7 7

Student Name: Ashley Putoni

Directory ID: bear4573

Instructions: Answer all questions. Show your reasoning where requested. For multiple-choice questions, circle one answer.

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1. Truth Table (6 points). Complete the truth table for the proposition $(p \wedge \neg q) \vee q$. Use T for true and F for false.

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D. If n is divisible by 4, then n is not even.

4. Short Answer (4 points). Define what it means for two propositions to be *logically equivalent.* Give one example of two logically equivalent propositions.

For all values/variables if we assign them T/F, then the expressions must be the same

5. Integer Proof (6 points). Prove that for every integer n, the integer $n^2 + n$ is even. Clearly state the definition of an even integer and justify your steps.

$n^2 + n = \frac{1}{2}(n+1) = 2k.$ ← Definition

$(n+1) = 2(2k)$

∴ $(n+1)$ is even so n^2+n is even ∎

CMSC250 Quiz 1

Discrete Mathematics | Instructor Saurav Patel

UID (9 digits): | I | A | M | B | A | T | M | A | N |

Student Name: Grace Adams

Directory ID: iambatman

Instructions: Answer all questions. Show your reasoning where requested. For multiple-choice questions, circle one answer.

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1. Truth Table (6 points). Complete the truth table for the proposition $(p \wedge \neg q) \vee q$. Use T for true and F for false.

p	q	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \vee q$
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T	F	T	T	T
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D. If n is divisible by 4, then n is not even.

I hate this question

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